

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

This assessment features a quick guide to the non-verbal reasoning tests, as well as an introduction to the 'missing square' questions your child might encounter. Then, there are two practice tests for your child to have a go at. We have included an answer guide too so that you can help your child to identify any misconceptions.

What skills does this practise?

Non-Verbal Reasoning

Problem-Solving

Missing Square

Further Activity Ideas and Suggestions

For more non-verbal reasoning practice, check out this [11+ Non-Verbal Reasoning Analogies Assessment Pack](#) and this [11+ Non-Verbal Reasoning Week of Workouts](#). You can encourage your child to get in some bite-size practice of all the 11+ skills with this set of [11+ Activity Mats](#).

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Non-Verbal Reasoning



The non-verbal reasoning test involves solving problems that are represented with pictures and diagrams. There are a number of different types of questions for the non-verbal reasoning test which could include:

- finding shapes that are like each other based on similarities (for example colours and number of sides);
- finding the odd one out in a group of pictures;
- identifying a shape which has been rotated;
- breaking the code in a sequence of images;
- spotting the missing shape in a sequence;
- identifying the nets of 3D shapes.



Here is an example of the type of question you may get in this section:



1. Which is the correct code for figure 4?

 M B	 R B	 M Q	 ?
1	2	3	4

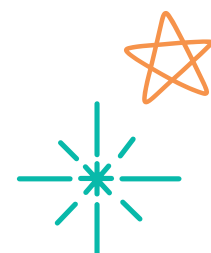
B	Q	B	R
Y	M	Q	Q
A	B	C	D



The answer would be D. The letter common to the first two shapes is B which represents the cross-hatching pattern they both have. Therefore Q represents the solid orange colour. The M is the letter representing the shape of the moon and it comes at the top of the box. Therefore the answer is D (R to represent the irregular dodecagon and the Q at the bottom to represent the solid orange colour).

The non-verbal reasoning test, for many children, can be very intimidating. The best way to prepare for this is through practice. Complete many different type of practice non-verbal reasoning tests with your child and discuss how each question is solved. This will give your child the experience needed to tackle the on-verbal reasoning tests.

For a thorough guide to all aspects of the 11+ test, try our **Survival Guide for Parents/Carers.**



11+ Non-Verbal Reasoning Practice Paper

Missing Square

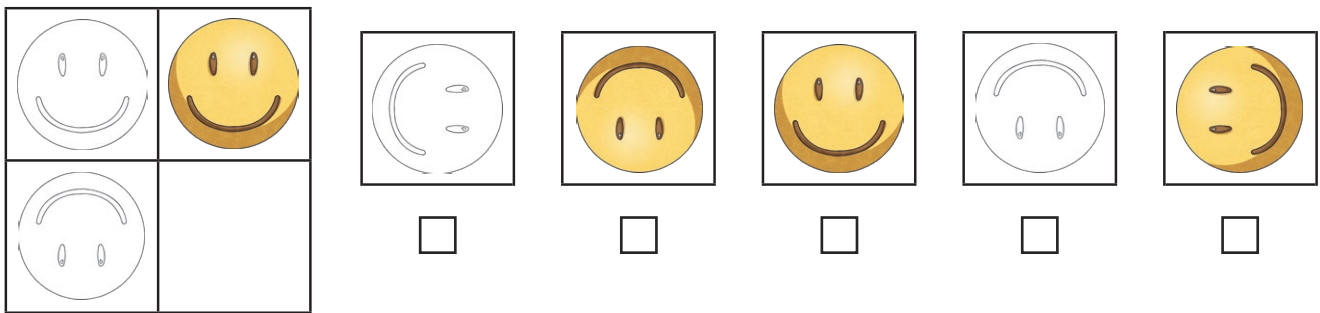
A common type of question that appears on many non-verbal reasoning papers is the **missing square** question.

This question type requires you to spot relationships or patterns between squares and identify which out of a selection will fill the gap.

Usually, you will be given a figure made up of either four or nine squares or tiles. One of the squares will be blank. You will also be given a selection of squares that could possibly fill that blank space.

Here is an example of a **missing square** question:

Tick the missing square.

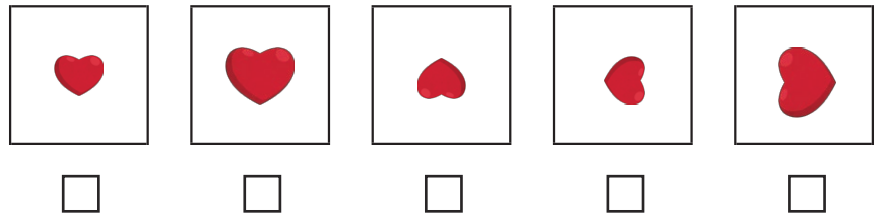
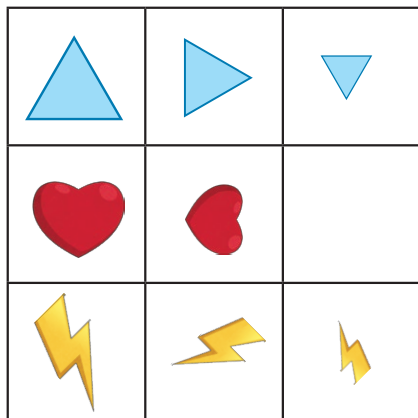


The answer is option 2. We can see that the bottom left square is a mirror image of the top left square. The top right square is the same as the top left but yellow. This means that the missing square must be a mirror image of the top right and must be yellow.



Here is a trickier example:

Tick the missing square.



The answer is option 3. Several things are happening in this figure. First, in each row, the shapes are decreasing in size from left to right. Second, the shape in each column is rotated 90° clockwise from the shape in the previous column. This means we are looking for a small heart which is rotated 180° from the first heart.

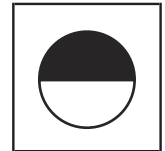
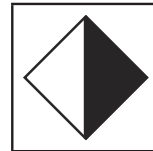
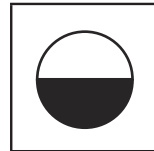
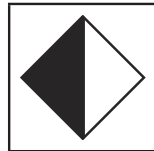
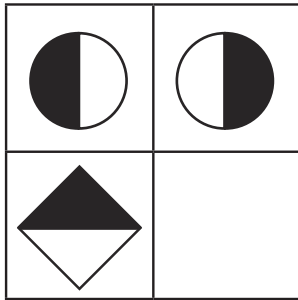


11+ Non-Verbal Reasoning Practice Paper

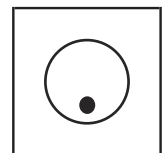
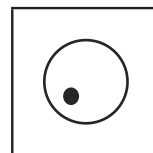
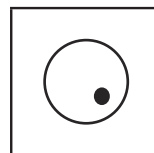
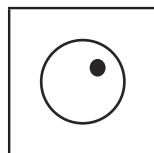
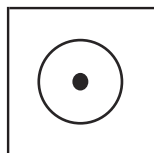
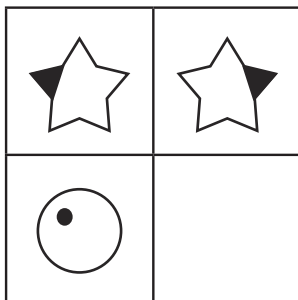
Missing Square: Test 1

Tick the missing square.

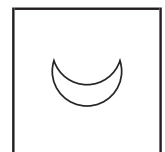
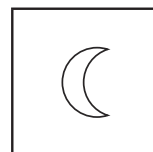
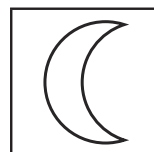
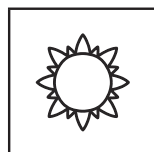
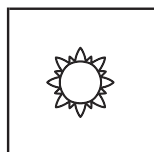
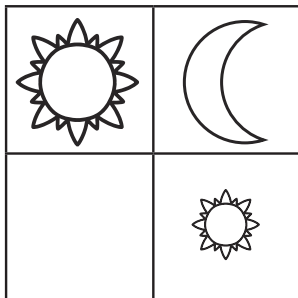
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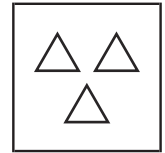
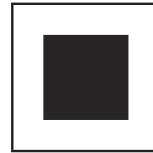
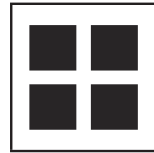
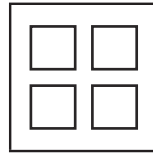
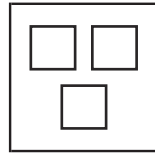
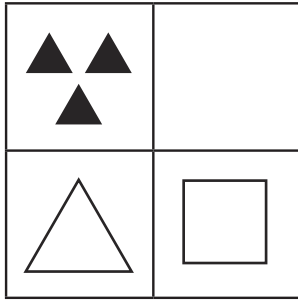
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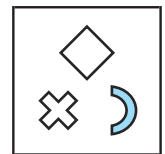
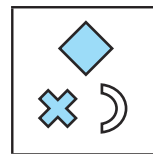
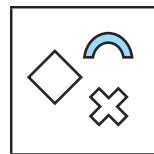
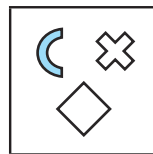
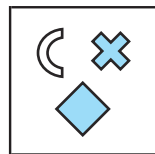
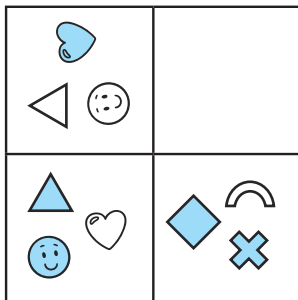
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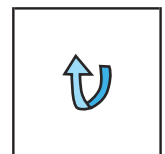
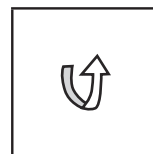
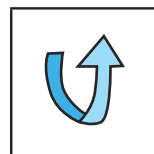
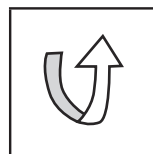
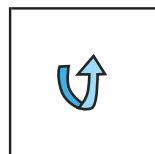
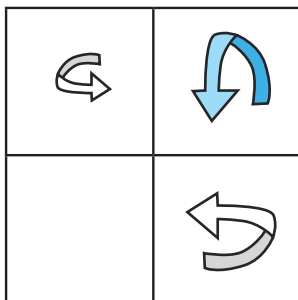
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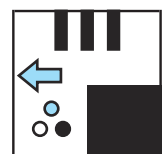
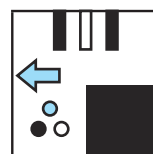
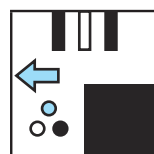
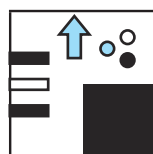
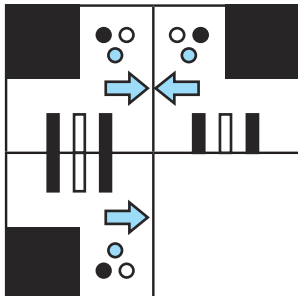
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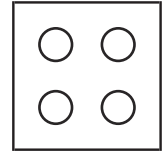
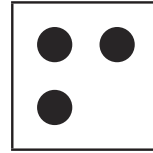
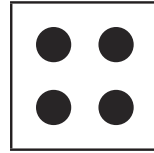
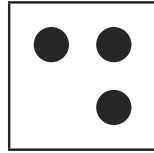
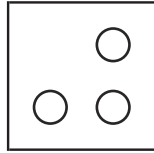
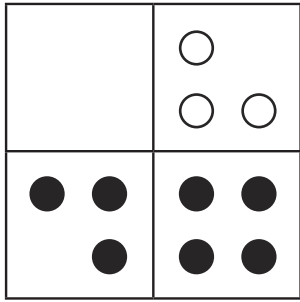
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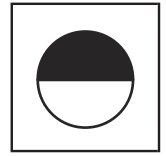
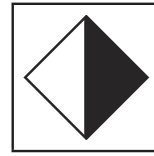
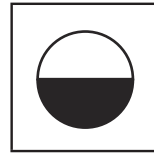
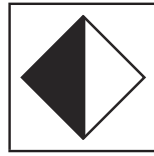
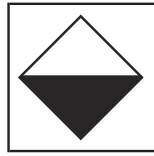
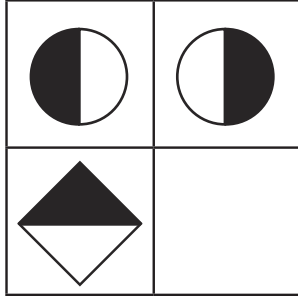
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11+ Non-Verbal Reasoning Practice Paper:

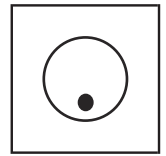
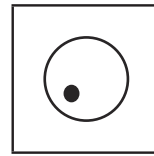
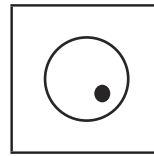
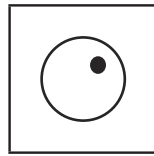
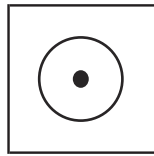
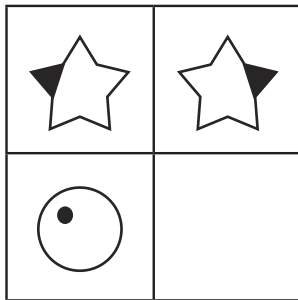
Missing Square: Test 1 Answers

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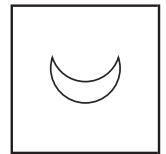
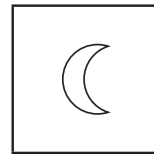
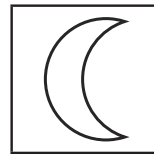
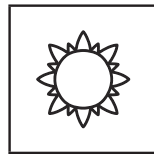
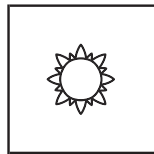
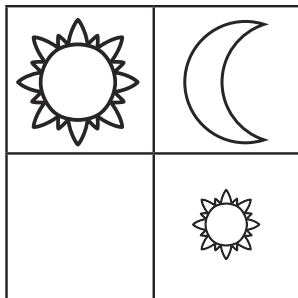
The missing square is option 1 because it is the same as the bottom left square but rotated 180°.

2.



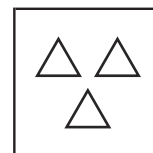
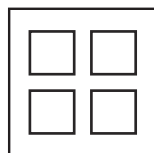
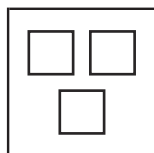
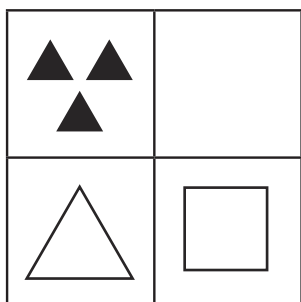
The missing square is option 2 because it is the mirror image of the bottom left square.

3.



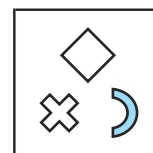
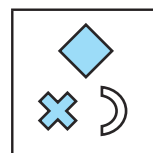
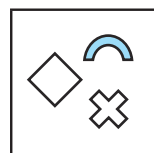
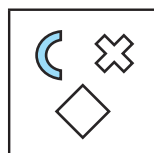
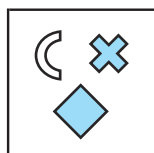
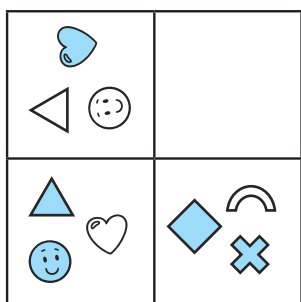
The missing square is option 4 because it is a smaller version of the top right square.

4.



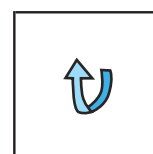
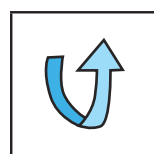
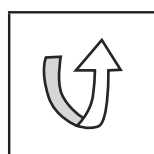
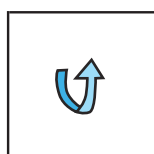
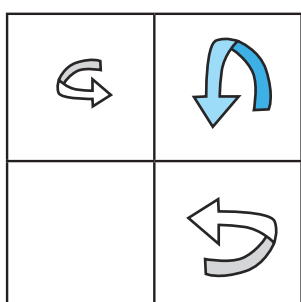
The missing square is option 3 because the top row contains smaller black versions of the shape in the bottom row. We can assume that the number of small shapes needs to match the number of sides on the shape, due to the relationship between the bottom left square and the top left square.

5.



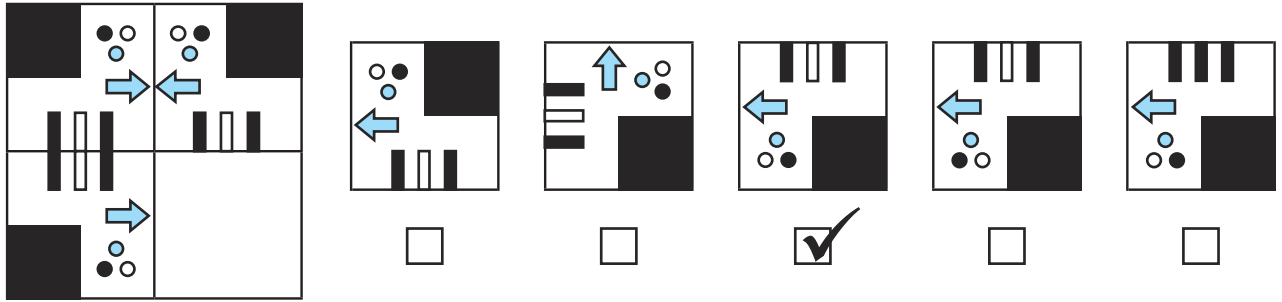
The missing square is option 2 because the top row figure has been rotated 90° clockwise to create the bottom row figure. The colours have also been inverted.

6.



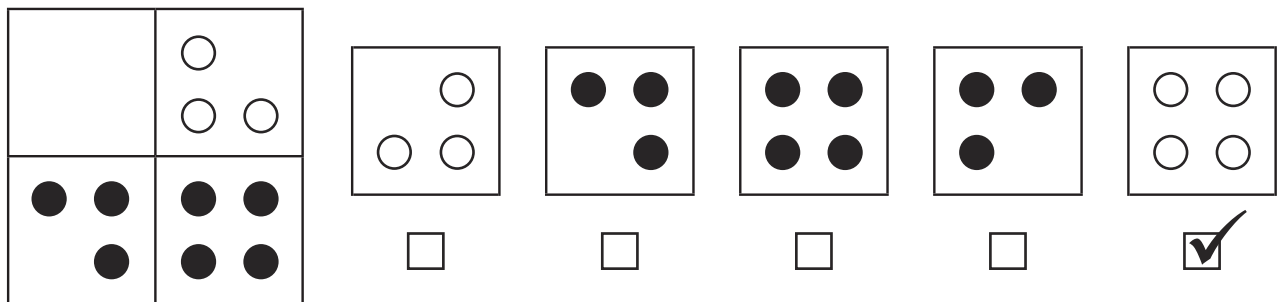
The answer is option 1 because the figure in the top right square needs to be rotated 180° and the size decreased.

7.



The answer is option 3 because it is a reflection of both the top right and the bottom left squares.

8.



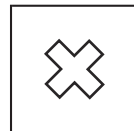
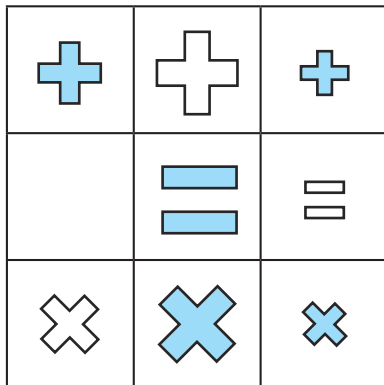
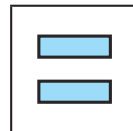
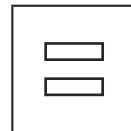
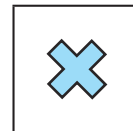
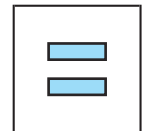
The answer is option 4 because it is a reflection of the bottom right square with the colour inverted.

11+ Non-Verbal Reasoning Practice Paper

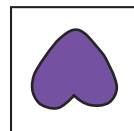
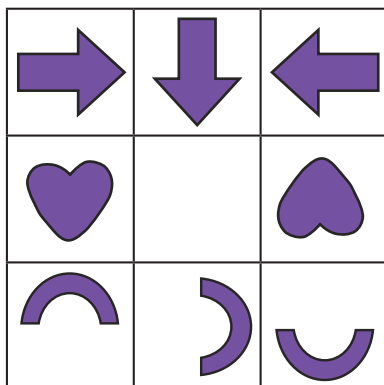
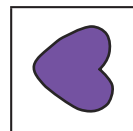
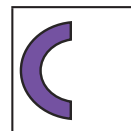
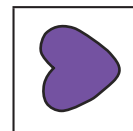
Missing Square: Test 2

Tick the missing square.

1.

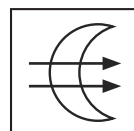
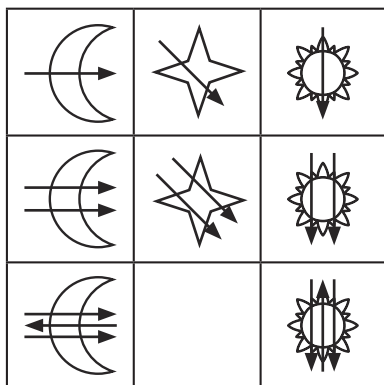

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2.


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3.


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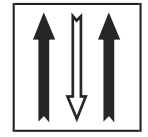
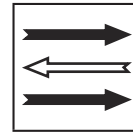
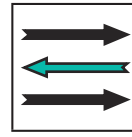
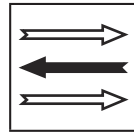
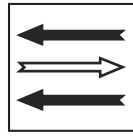
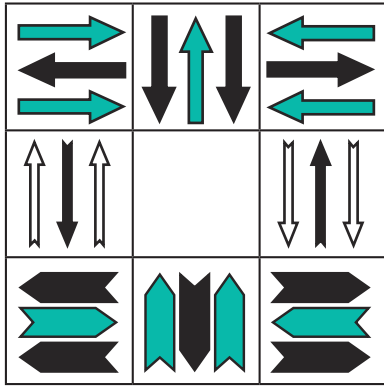
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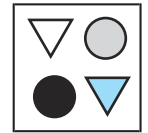
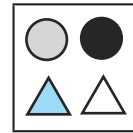
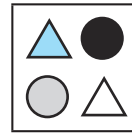
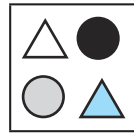
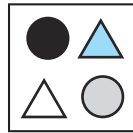
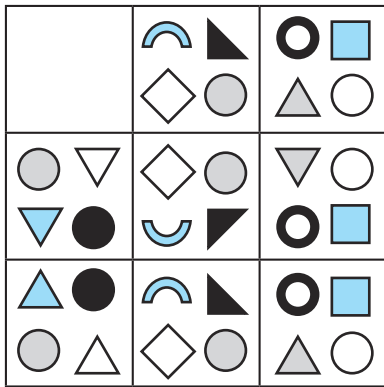
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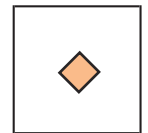
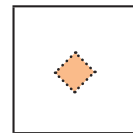
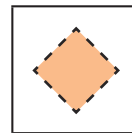
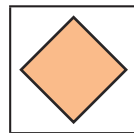
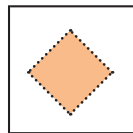
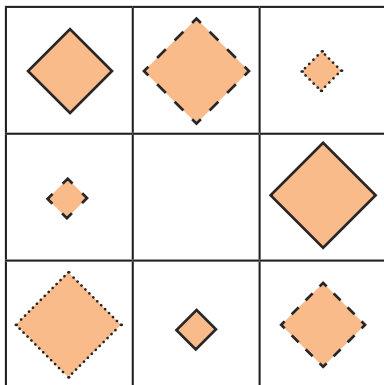
4.



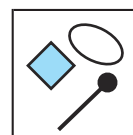
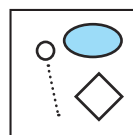
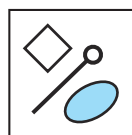
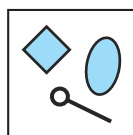
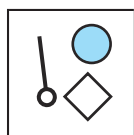
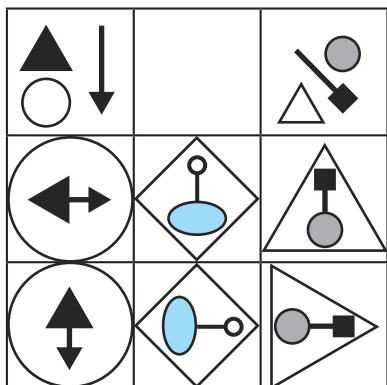
5.



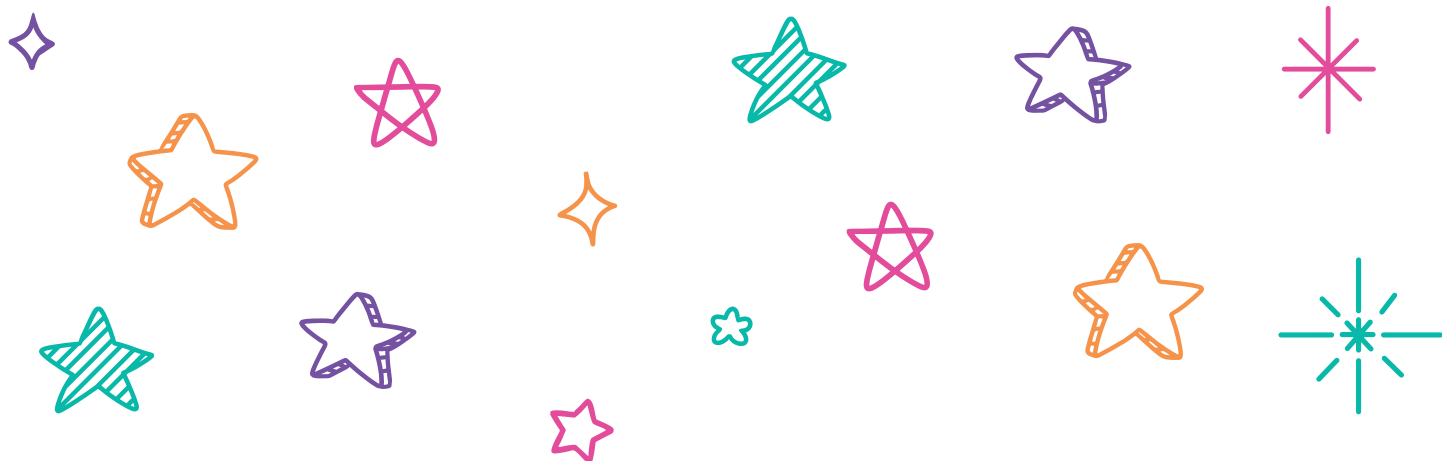
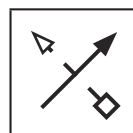
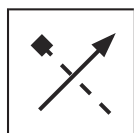
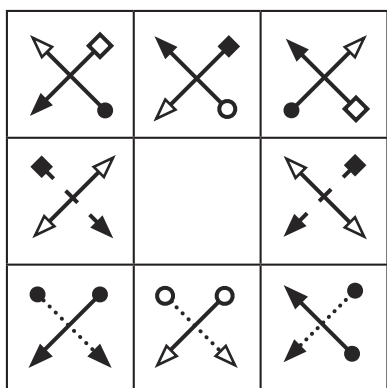
6.



7.



8.

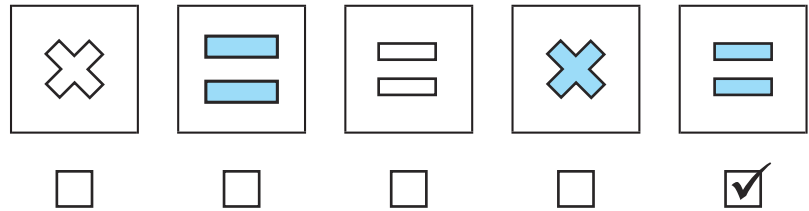
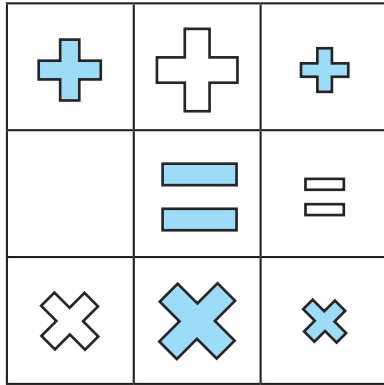


11+ Non-Verbal Reasoning Practice Paper:

Missing Square: Test 2 Answers

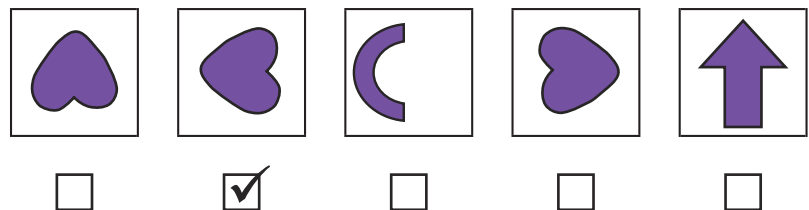
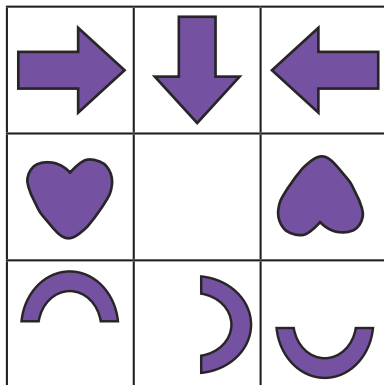
Tick the missing square.

1.



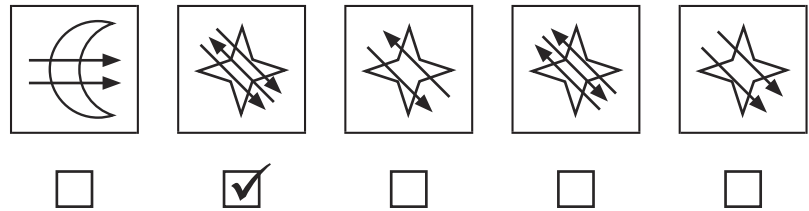
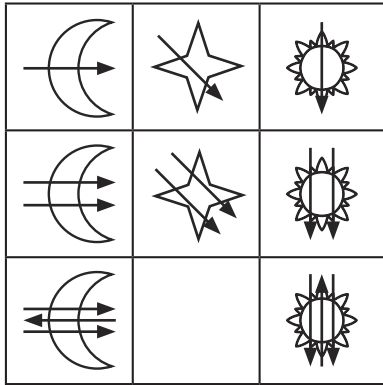
The answer is option 5 because the first column features medium size shapes. It must be a blue shape because there is only one white shape in each column and row.

2.



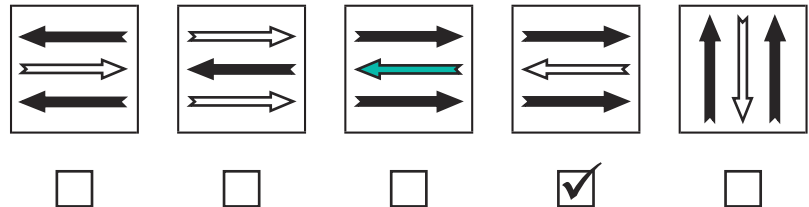
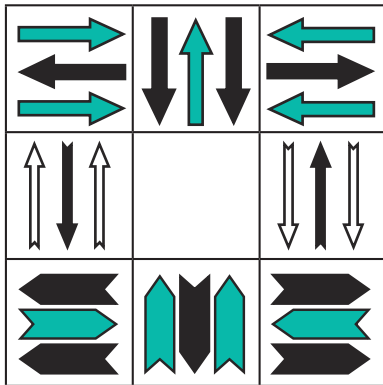
The answer is option 2 because the shapes are rotating 90° clockwise in each row.

3.



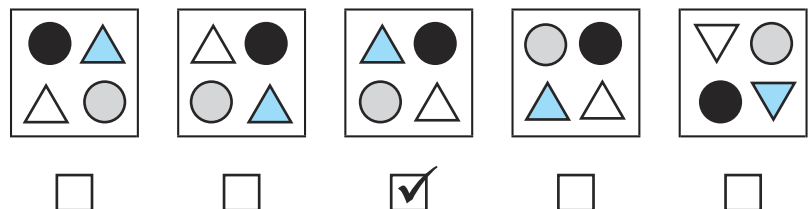
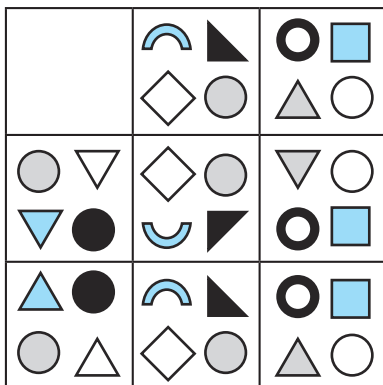
The answer is option 2 because the shape must be the same as the others in the middle column, but an additional arrow facing the opposite direction is added.

4.



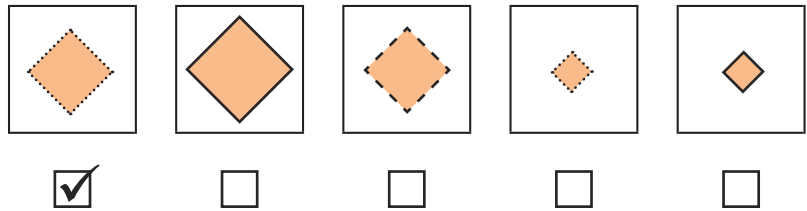
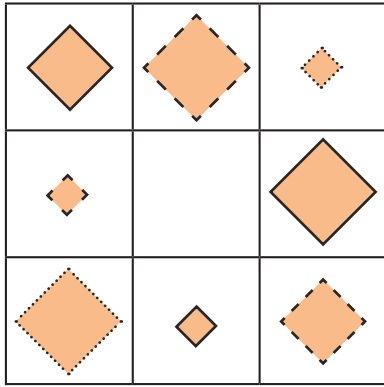
The answer is option 4 because the figure in the first column is rotated clockwise by 90° and the colours are inverted.

5.



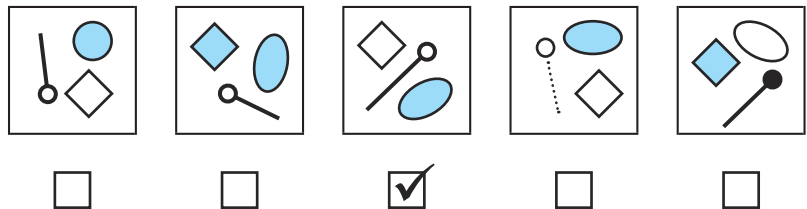
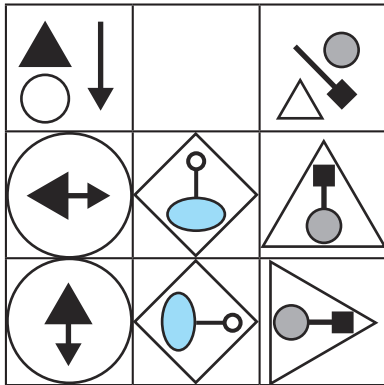
The answer is option 3 because the figures in the top row are a reflection of the figures in the middle row.

6.



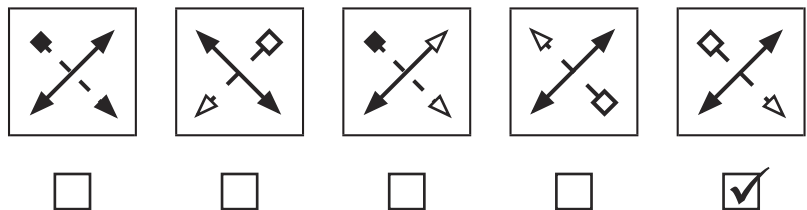
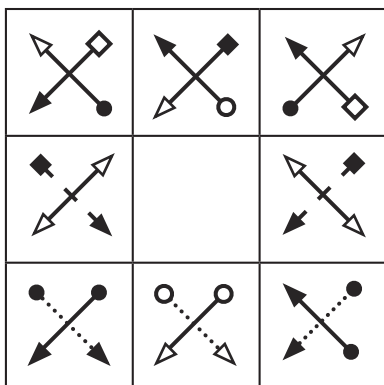
The answer is option 1 because there is a small, medium and large variety of each of the squares, so we are missing a medium-sized, dotted shape.

7.



The answer is option 3 because the shapes in the middle of the rows in the second column are made up of the same components – a white rhombus, a blue semi circle and a solid line with a circle on the end.

8.



The answer is option 5 because in the middle column, the figures from the first column have stayed in the same position but the colours have been inverted.

Disclaimers: We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.